Inspecting Boston Rental Price Dynamics: A Comparative Analysis of Short-Term and Long-term Pricing

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Abstract

The purpose of this project is to explore how rental prices are set in short-term and long-term markets. Airbnb datasets and housing market data sets are compared to observe key differences in pricing drivers. Machine learning models were used to predict price and identify the features that influence price. Two models were used: a Linear regression model (baseline) and a Random forest model. The results show that Airbnb pricing is influenced primarily by listing level features like room type, capacity, and local. While long-term rentals pricing is far more predictable and attached to macro factors like location and time. This comparison underlines the difference between short and long -term rentals, and shows the challenges of trying to use their pricing patterns in tandem.

1 Introduction

This project aims to explore the differences between short and long term rentals in the Greater Boston area. Short term rentals are evaluated using a comprehensive Airbnb dataset, while long-term rentals are represented by Zillow market data. The drivers of price in both cases is affected by different factors.

Short-term rental platforms record data at the listing level; meaning price is influenced by factors like room type, location, capacity, and customer signals. Long-term rental information is observed at the macro level, this means pricing stability and factors of influence like location, and time on market.

The purpose of this project is to understands if the pricing patterns learned from the Airbnb dataset can assist in explaining long-term rental markets. To achieve this, the focus was placed on identifying key differences between both markets. Machine learning models were trained on both datasets to compare performance and feature importance.

The results showed that where Airbnb is influenced by a granular combination of listing features, long term rental prices are far easier to predict and mostly depended on location and temporal factors.

2 Data

This project is centered around two datasets that highlight pricing dynamics in both short and long-term rental markets. Both datasets were sourced from Kaggle. The first is a collection of Airbnb listings from the Boston area detailing features of the rental properties. In order to prepare this dataset for modeling, non-numeric fields were cleaned and converted to usable formats, missing values were handled using median imputation, and unimportant or redundant features were dropped. Since the listing prices were so skewed, a log transformation was also applied to reduce overall variance.

The second dataset has aggregated long term rental data across towns surrounding the Greater Boston Area. This dataset had a leaner feature set which included years on market, location, and rental/bedroom count. For the purposes of this project houses were filtered to only include two bedroom houses located in Boston; this is to allow consistency in the target for analysis. As opposed to its Airbnb counterpart this dataset was macro focused, emphasizing on overall market trends as opposed to individual listings. This dataset required very little pre-processing compared to the other. Target entries with missing values were removed, and categorical features (town/county) were hot encoded for ease of use with the ML models.

Put together these two dataset enable the observations of micro pricing vs macro pricing as seen in short vs long term rental markets.

3 Methods

The Boston rental analysis project centers around two primary machine learning approaches: Linear regression, and Random Forest regression.

Linear regression serves as a baseline model to establish a simple yet viable relationship between input features and rental prices. This model assumes a linear relationship between variables and provides a path to evaluate the relationship between base features and pricing. While Linear regression is simple to interpret, it is rather lacking in its ability to capture more interesting and noteworthy patterns in the data.

In order to address this lack of completeness, a Random forest regression was used to test the data against a non-linear model. Random forest is an ensemble method, this is to say it uses a collection of weak learners to create one stronger learner that outputs predictions with a higher accuracy level. This helps identify any nonlinear relationships that the regression model failed to capture.

Both models were applied to the Airbnb dataset to ensure the more complex model actually captured nuances missed by the simpler regression model. For the long-term rental dataset only the random forest model was applied; this is because the purpose of this dataset was to identify the dominant drivers at a macro level.

4 Experimental Setup

All of the experiments were ran using Python libraries. Both data sets were treated in separate notebooks but followed the same procedural steps. The libraries used in this project can be seen in the table below

Library	Description
pandas	Used for data loading, cleaning, and manipulation of tabular datasets.
numpy	Numerical operations and transformations, including log scaling.
scikit-learn	Machine learning models, preprocessing pipelines, and evaluation metrics.
seaborn	Statistical data visualization.
matplotlib	Visualizations, including scatter plots and bar charts.

Table 1: Libraries used for data processing, modeling, and visualization.

For the Airbnb dataset, the target variable was price and it was log-transformed log(1+x) in order to reduce skew. The feature set was tightened significantly to include only the key features in the table

below. Features were converted to the appropriate type and in the case of categorical variables, they were hot encoded.

Feature	Reason for Inclusion
Room Type	Tells us Whether the listing is an entire home or shared space
Bedrooms, Bathrooms, Accommodates	Represent property size and capacity (how many guests a listing can support.)
Latitude, Longitude	Encode geographic location, captures neighborhood desirability and proximity to key areas.
Availability (365)	how often a listing is available, acting as a proxy for demand and pricing strategy.
Number of Reviews	Listing popularity and demand over time.
Review Scores	Represents perceived quality, (potentially influences pricing decisions.)
Minimum Nights	Booking constraints that may affect pricing behavior.

Table 2: Selected Airbnb features and their roles in predicting listing prices.

The Airbnb dataset was split into training and testing using the standard 80/20 split with a fixed random state (28) for reproducibility. For the Airbnb dataset categorical features were encoded using one-hot encoding, and numerical features were used as is. For consistency pre-processing steps stayed the same across both tested models.

The Boston housing dataset underwent minimal processing, but was filtered to only include two-bedroom rental prices to create consistency in the data. The features used for model were year, town, and county. Records with missing values in the target variable where removed , and categorical variables were hot encoded. A random Forest model containing 100 trees was trained using the exact same 80/20 split and evaluation tooling as the Airbnb set. The same set up was used to compare the difference between long term and short term modeling side by side.

5 Results

Both models performances were evaluated with Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and the coefficient of determination ($R\hat{2}$). These metrics were used to evaluate prediction accuracy and model fit.

5.1 Airbnb Dataset Results

The Airbnb dataset being our primary focus, it was put through both a linear regression and random forest model. The linear regression baseline resulted in an R^2 score of about .64, with a MAE of around $49 and a RMSE of around $100 when converted back to dollar values. These numbers show that the model was able to showcase a large portion of the variance in listing prices but still produced decently large prediction errors.

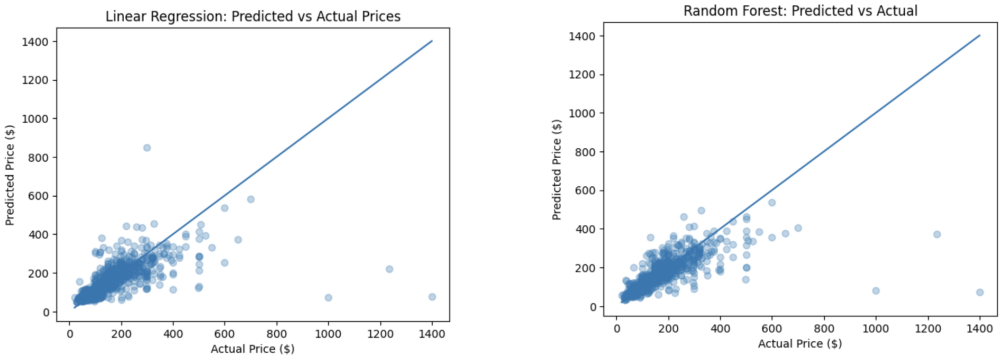
The random forest model significantly improved performance. It achieved a R^2 score of about .71, with a MAE of $43 and an RMSE of $92. This improvement means that the regression model failed to capture some of the non-linear relationships between features, which evidently are key to pricing variation in airbnb listings.

Metric	Linear Regression	Random Forest	Difference
R^2	0.642	0.710	+0.068
MAE ($)	49.20	43.29	-5.91
RMSE ($)	100.15	91.64	-8.51

Table 3: Performance comparison between Linear Regression and Random Forest models on the Airbnb dataset.

The graphs below highlight both model's predicted vs actual price, making the differences between the models clearer. Both models perform better when predicting pricing for lower to mid range

listings as indicated by clusters near the diagonal line in those ranges. The key difference is that the linear regression model shows a much wider spread around the diagonal, even more so as the price increases, this is a reflection of its larger prediction error value. The random forest model outputs a much tighter clustering of points in the provided data, it improves accuracy and allows us to observe nonlinear relationships in the data. In spite of the improvements seen by the RFM, both models fall short when it comes to predicting for high-end listings. As listing price increases they both consistently under price by a significant margin. This means neither model are capable of identifying the factors that drive luxury rental pricing. Overall we can conclude that incorporating a nonlinear model greatly improves performance for typical listings, but difficulties remain when modeling high-end property pricing. This is most likely caused by missing luxury features and a greater variance in pricing behavior as we reach market ceiling.



(a) Linear Regression

(b) Random Forest

Figure 1: Predicted vs. actual prices for Linear Regression and Random Forest models.

5.2 Long-term Dataset Results

For the long-term housing dataset, a RFM was trained to predict our two-bedroom rental prices. The model performed very well, achieving a $R\hat{2}$ core of around .97, a MAE of around $28 and an RMSE of $71. These are really strong results, which leads to believe the model is highly capable of observing long term rental prices with a relatively small average prediction error.

Compared to the airbnb dataset, this model had a significantly higher predictive accuracy; this suggests that long-term rental prediction is much more stable and structured than its short-term counterpart.

The figure below shows us the average two-bedroom rental pricing over time, it clues us into why the model's accuracy was much higher on this dataset. The trend is consistently moving upward with a notable increase in growth around 2017. This very steady growth pattern explains the RFM's performance. Unlike the airbnb dataset, this one follows a predictable temporal trend making it a perfect candidate for predictive modeling.

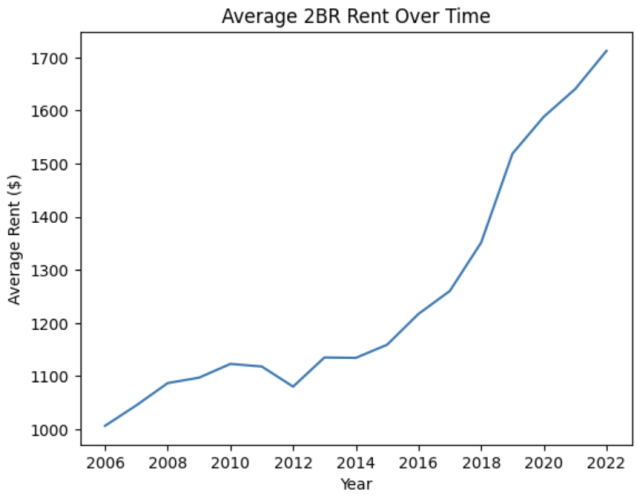


Figure 2: Average two-bedroom rent over time in the housing dataset.

6 Analysis

The resulting outcomes of the models show a clear difference in predictability of pricing between the airbnb and the GBA housing datasets. For airbnb, the random forest model had an R^2 of around.71, where the long-term housing data reached an R^2 of .97. This gap can also be observed in the error metrics with airbnb predictions showing much larger deviations compared to the minimal errors in the housing model. Taking a closer look at the predicted vs actual for the airbnb dataset, we can see both the linear regression model and random forest model did decently well in the low to mid price range ($50-$300 per night). But as previously mentioned both models consistently under-priced luxury rentals, with some of the errors getting as large as actual prices around $500. The random forest reduced the spread compared to the linear model but this trouble identifying high priced rentals stayed consistent implying the limitation is not model specific.

A look into feature importance allows us to explain this pattern. In the airbnb models, room type (especially shared space vs entire home rentals) was by far the most important feature, the following most impaction features being latitude and longitude. This tells us that the listing type and location are the dominant drivers for price. Unfortunately the remaining features contribute very little to the price, this means that a large portion of variation is not observable via the provided data. Conversely, the GBA dataset had a much stronger structure, the time series plot showed a steady increase from around $1000 in the mid 2000s all the way to $1700 by 2022. Feature importance reflects this observation with year and location absolutely dominating in price influence. Because these variables are responsible for most of the variation in the data, the model is able to make very accurate predictions.

Feature	**Importance (%)**
Room Type (Entire Home/Apt)	45.79
Latitude	10.37
Longitude	10.18
Availability (365)	5.85
Bathrooms	5.74
Accommodates	4.14
Bedrooms	3.64
Number of Reviews	3.28
Minimum Nights	2.87
Review Scores	2.41

Table 4: Feature importance from the Random Forest model for Airbnb pricing.

Putting these results together, we can determine that the primary limitation with the Airbnb model is not the algorithm but the data itself. Airbnb pricing seems to be influenced by additional factors that do not appear in the dataset. Things like amenities, listing quality, or host strategies probably come into play. Whereas the long term housing market is very stable in its pattern and driven almost entirely by year and location.

7 Limitations

This project started with the goal of using Airbnb pricing data to predict long-term rental pricing based on property features. While the models were able to prove a meaningful relationship between price and features, the results were not strong enough to support the use of Airbnb data alone to extrapolate long term rental value. Based on the results short-term pricing is far too fluctuant because it is affected by a wide breadth of variables. This suggests that Airbnb data cannot be used as a work around for Zillow's well guarded data on determining long-term rental prices.

8 Conclusion

The purpose of this project was to compare the pricing dynamics of short-term and long-term rental markets in the Greater Boston Area using machine learning techniques. Through the process a very clear distinction became apparent between the two markets. Short-term rental pricing is driven by very granular and highly variant listing level features. This resulted in okay predictive performance even with more complex modeling techniques. The long-term pricing market acted in contrast by following a very structured and predictable pattern primarily dictated by time and locations. This projects biggest take away is that while both markets are tied to property pricing, they operate in different lanes. Short-term rentals are dynamic, heavily influenced by demand fluctuation, listing strategies, and unrecorded factors. Conversely, long-term rentals behave closer to traditional economics where macro-trends dictate the market. This study showcases the idea that you can't force data to answer a question it can't answer. Data context is just as important as modeling techniques and model performance is not entirely dependent on algorithm selection, it also requires the provided features to operate as drivers for the target. Future work would see focus on closing the feature gap, this could include using a richer feature set of Airbnb attributes, or even combining both datasets into a unified set that reflects both micro-level variance and macro-level stability. Without a solution to the disconnect in the data, attempts to garner insights from one market to the other will not prove particularly useful.